

ASGA: Attention-based Sparse Global Attack to Video Action Recognition

Zeyu Zhao, Ke Xu, Tanfeng Sun, *Senior Member, IEEE*, Xinghao Jiang, *Senior Member, IEEE*

Abstract—Recent years have witnessed significant progress in adversarial attacks and defenses for video content understanding. Most methods craft perturbations on an independent sample basis without considering inter-frame consistency and temporal context difference. When the target model changes the frame sampling method, the success rate of the original video adversarial attack method will significantly decline. If attackers attack every frame individually, the computational cost will be too high, and the attack success rate will not improve substantially. This paper proposed a novel attention-based sparse universal attack that enhances perturbation inter-frame transferability on videos. Firstly, the key frame sequence is selected according to the feature similarity between frames. The attention map of the target model is then calculated on keyframes to identify semantically critical regions for perturbation generation. Temporal translation is further applied to improve inter-frame perturbation transferability. The integral frame perturbations are finally replicated to non-keyframes. The essential advantage of our proposed approach is that the adversarial perturbation only needs to be computed once on a keyframe sample yet can achieve universal attacks on the entire video sequence. Our proposed attack method incurs the exact computational cost of crafting perturbations for individual samples. Moreover, integrating attention maps encourages sparsity in the optimized adversarial perturbation. This paper evaluates global attack capability under two random sampling strategies: 1) randomly sampling N frames from all frames, and 2) randomly picking a starting frame and selecting $N-1$ consecutive frames after it. Experiments on multiple models and datasets demonstrate the transferability of our approach under various frame sampling methods.

Index Terms—Universal Video Adversarial Attack, Attention Mask, Sparse Attack

I. INTRODUCTION

Recently, the vulnerability of DNN models has attracted significant research. By adding some imperceptible adversarial perturbations to clean samples, although the semantic information of the samples remains unchanged, the well-performance model's outputs become entirely different [1], [2]. In many fields, there is a severe threat from adversarial attacks, such as face recognition, video analysis, autonomous driving, etc. As a result, adversarial examples have attracted numerous research attention in recent years.

There has been significant research on adversarial video attacks. Video adversarial attacks oriented toward the goal of sparse attack will confine adversarial perturbations to specific frames and specific regions through different optimization strategies [3] [4] [5] [6] [7]. On the other hand, adversarial example attack methods targeting robustness will focus on enhancing specific robustness against current common image and video processing methods [8], such as lossy coding compression [9] [10], adversarial purification, and adversarial

detection [11]. All the video adversarial attack methods mentioned performed well in the scenario they set. However, they need to catch up on a crucial scenario: the reduction of attack success rate caused by changes in video keyframe sampling methods.

In modern video action recognition models, the input is a fixed-length frame sequence called an input sample. When recognizing video samples, they are first segmented into sections with a length that matches the model's input. Each segment is then identified as an input sample, and finally, the recognition inference result for the entire video is obtained through voting or segment output. There are many ways to segment video, also known as frame sampling methods. The most common ones include continuously taking N frames, taking N frames every other frame, etc. Changes in the sampling method can lead to changes in the input samples. Therefore, when conducting adversarial attacks on videos, if only one segment of the video is attacked, then when the frame sampling method changes, not all adversarial frames participate in the inference, which can lead to attack failure. If attacks are conducted on each segment, not only the computational cost is extremely high, but also the attack effectiveness will decrease due to the lack of consideration for the transferability of different sample perturbations within the other frames.

To address above problems, this paper proposed an attention-based sparse global video adversarial attack. The attack process is shown in Figure 1. Our attack method includes keyframe selection, attention-based sparse attack, and perturbation diffusion. An ImageNet-1k pretrained image classification model is used in the keyframe selection step to extract frame features. Then, all frames are divided into N groups by calculating the cosine similarity between adjacent frames, where N is the number of input frames needed by the target model. In each group, find the frame with the highest similarity to the rest of the frames as a representative frame for that group and use the N representative frames as selected keyframes for the attack. In the attention-based sparse attack step, an attention mask is generated to guarantee adversarial perturbation sparsity. Then, a temporal recurrent translation is used to enhance the inter-frame adversarial transferability. At last, keyframe adversarial perturbation is caused and spreads to other non-keyframes to achieve a global attack effect.

In this paper, our contributions are as follows:

- 1) In this paper, an attention-based sparse global attack is proposed to overcome the frames sampling methods uncertainty.
- 2) A new measure of the performance of video adversarial attacks is proposed: the global attack success rate

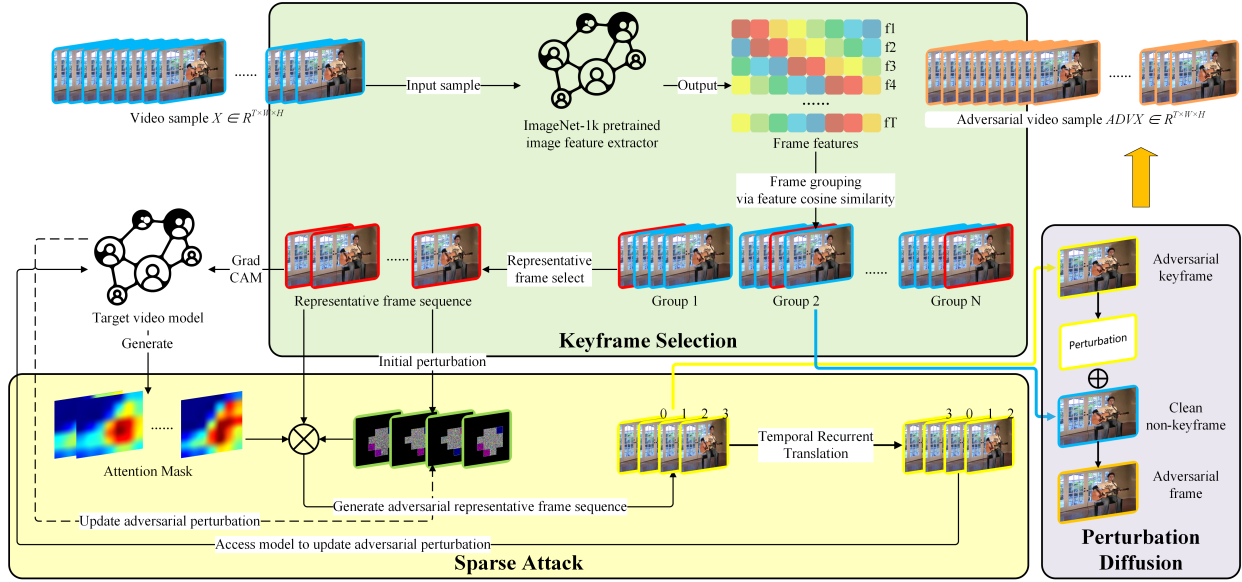


Fig. 1: The global video adversarial attack method proposed in this paper.

(GASR) of adversarial attacks.

- 3) Extensive experiments demonstrate that our proposed method performs well under a global attack setting.

II. RELATED WORK

A. Video Action Recognition

Video action recognition is a crucial task in video content understanding. Current state-of-the-art approaches can be categorized as convolutional neural network (CNN) based models and Vision Transformer (ViT) based models. CNN models utilize 2D or 3D convolutional networks [12]–[14] for spatio-temporal feature extraction. Through multiple convolutional layers, input frames are mapped to fixed-length vector representations for final classification. ViT models adopt a Transformer architecture [15], dividing input frames into patches, embedding them, and connecting terms through multi-headed self-attention [16], [17]. While computationally intensive, ViT models achieve superior performance on large-scale benchmarks. In this work, CNN-based and ViT-based video action recognition models demonstrate the effectiveness of changing sampling methods. Extensive experiments validate the existence of diminished global attack success rates across model architectures.

B. Adversarial Attack

While deep neural networks achieve excellent performance in multiple areas of computer vision, the appearance of adversarial examples makes people think more about DNN's robustness and safety. By adding minor disturbances that are invisible to the human eyes in a clean sample. The target model cannot classify the samples correctly; the objective of an adversarial attack can be formulated as Equation 1, where $f(x)$ denotes the DNN model, x is the input sample, δ is added adversarial noise, y is truth label, ϵ is the upper limit of attack amplitude.

$$f(x + \delta) \neq y, |\delta| < \epsilon \quad (1)$$

There are several methods for generating adversarial image examples. Fast gradient sign method (FGSM) [18] uses the gradient information to update the input examples and attack the neural network successfully. Projected gradient descent (PGD) [19] algorithm also uses gradient information to update clean samples until the samples can be classified wrong. Unlike FGSM, PGD considers the nonlinear feature of CNN, and it uses more but smaller update steps to generate adversarial examples. Besides, the C&W algorithm [20] is an optimization-based adversarial attack method. This algorithm regards classification loss and disturbance amplitude as optimization targets and reduces the range of disturbance as much as possible while attacking the model. The adversarial examples generated by this method have good concealment in vision.

The generation of video adversarial examples is very similar to adversarial image examples. In Sparse Adversarial Perturbations for Videos [3], the researchers attacked the CNN+RNN model and proposed a kind of optimization-based adversarial attack. In MSANet Attack [21], researchers have incorporated attention mechanisms at the video, frame, and feature levels to reduce perturbation magnitudes while maintaining success rates significantly.

The above methods can generate adversarial examples with a high attack success rate, but none consider the impact of frame sampling methods. Due to fixed input frame counts, video action recognition models operate on clipped subsets of complete sequences during inference. Varying the frame sampling strategy impacts recognition and accuracy.

Under the scenario of changing sampling mode, the attack's success rate decreases sharply in different attack modes. We extend prior methods for fair benchmarking to enable global attacks under uniform sampling. First, attacks are generated typically per the original methodology on an input sample. The adversarial perturbation is then replicated across all frames, which allows comparison under similar compute constraints as our proposed approach.

III. METHODOLOGY

A. Symbol Definition

The main symbols used in this paper and their definitions are presented in Table I.

TABLE I: Symbols used in the this work

Symbol	Definition
x	Single video sample.
y	The true label of x .
N	Required frames length of video action recognition model.
x_i	Single video clip sample with a specific length for model input.
$S(x, N)$	Sample N frames from the video sample x .
$f(x_i)$	Video action recognition model, ensuring that $f(x_i) = y$
δ	Adversarial perturbation generated by adversarial attacks.
ϵ	The maximum absolute magnitude boundary of adversarial perturbations.
x_{adv}	Video adversarial sample.
$RS(x, N)$	Randomly sample N frames from the video sample x .
k	The number of random samplings during global attack testing.
$e(x_i)$	Imagenet1k-pretrained image feature extractor.
g_i	Gradient information of x_i .

B. Problem Formulation

Given a sample $x \in X \subset R^{C \times T \times W \times H}$ with its ground truth label $y \in Y = \{0, 1, 2, \dots, K - 1\}$, where C, T, W, H denote the color channels, number of frames, width, and height, K represents the number of classes. For a given video action recognition model $f(\cdot)$, selecting a certain number of frames from the video and combining them into a model-acceptable input format for prediction is standard practice. Denote $x_i = S(x, N) = \{x_{i_0}, x_{i_1}, \dots, x_{i_N}\} \subset R^{C \times N \times W \times H}$ as input sample, where N denotes the number of input frames that the model can accept, $S(x, N)$ denotes the selection of N frames from samples x , and $N \leq T$, x_{i_n} represents n_{th} frame in x_i . Thus, the prediction process of the model for the sample can be expressed as $f(x_i) = y$. For the previous adversarial attacks, the goal of their attacks is to make the model's inference about the sample wrong after adding a perturbation [22] [23], as shown in Equation 2. Where δ_i denotes the adversarial perturbation generated for a sequence x_i , ϵ denotes the parameter that constrains the attack magnitude, generally using L_2 or L_∞ norm.

$$f(x_i + \delta_i) \neq y, \quad |\delta_i| < \epsilon \quad (2)$$

However, when the target network decides to use a different sampling method, the effectiveness of the adversarial attack decreases dramatically if a sufficient number of adversarial frames are not selected. Thus, a global video adversarial attack is proposed; the attack's target is shown in Equation 3. Where x_{adv} represents a sequence of adversarial frames obtained by sampling from x that has undergone a global attack, δ_{global} denotes the global adversarial perturbation added in sample x .

$$\begin{aligned} \forall x_{adv} \in \{x_{adv} | x_{adv} = S(x + \delta_{global}, N)\} \\ f(x_{adv}) \neq y, |\delta_{global}| < \epsilon \end{aligned} \quad (3)$$

For global attacks, when an attacker attacks a video sample, the attack is successful regardless of the sampling method taken by the target model. Considering that testing the attack for all sampling methods in a global adversarial attack is impossible, an index that can effectively test different sampling methods, namely the global attack success rate (GASR), should be proposed. By randomly sampling k times among the samples x that have been attacked to obtain k adversarial sequences, testing these k sequences, and calculating the average attack success rate, it is regarded as the global attack success rate for a sample x . The global adversarial attack success rate is defined by Equation 4, where $RS(x, N)$ means that N frames are randomly selected from sample x and combined into a prescribed format. This paper adopted two random sampling methods: randomly sample N frames and randomly choose a first frame, then consecutively sample $N - 1$ frames.

Consider two attack scenarios and sampling strategies for evaluating adversarial potency. First, if the attacker perturbs m length- N segments across the original frames, testing on random N -frame subsets risks sampling correlated pieces and underestimating global impact. Random start-frame selection is more appropriate. Conversely, if the attacker perturbs only N seed frames and propagates to all frames, random N -frame selection may resemble uniform sampling, fully capturing the attack trace. Random start frames would then be preferred.

$$GASR(f, x, y, k, \delta) = \frac{1}{k} \sum_{m=0}^k \begin{cases} 1, f(RS(x + \delta, N)) \neq y \\ 0, f(RS(x + \delta, N)) = y \end{cases} \quad (4)$$

Therefore, the optimization objective of the global video attack is given by Equation 5.

$$\begin{aligned} \underset{\delta_{global}}{argmax} (GASR(f, x, y, k, \delta_{global}) - |\delta_{global}|) \\ \lambda > 0, \quad |\delta_{global}| < \epsilon \end{aligned} \quad (5)$$

C. Attention-based Sparse Global Attack

To achieve the global attack, performing attack on each frame in the video is necessary. Nevertheless, this is extremely difficult for three main reasons:

- 1) A video contains many frames, if the perturbation on each frame is to be calculated separately, it requires considerable computational overhead.
- 2) If each set of video frame sequences is perturbed separately, it is not easy to guarantee that the recombined frame sequences can still be successfully attacked.
- 3) Suppose the attack is performed on every position of the frames in the video. In that case, it will lead to a very obvious attack perturbation and lose the meaning of the adversarial sample.

To solve the above problems, a sparse global video adversarial attack method based on an attention mechanism is proposed. The effect of global adversarial attacks can be achieved by computing adversarial perturbations in only one set of video frame sequences and replicating these adversarial perturbations in the remaining frames.

1) *keyframe Select*: In practice, the number of frames in a video often far exceeds the number required as input to recognition models. Action classifiers commonly subsample frames via interval or multi-pass sampling strategies to enable efficient inference. In such cases, perturbing every frame is prohibitively expensive and needs higher generation efficiency. Moreover, it cannot guarantee consistent correlations between perturbations added in different input batches, leading to a poor global attack success rate. To address this problem, an approach that crafts adversarial perturbations on only the video sample's most representative N keyframes is proposed. Then, the N frames are called keyframes, and the perturbation on the keyframes is called key perturbation, which is represented as δ_{key} . The perturbation generated on these keyframes exhibits greater transferability to other non-keyframes during model inference.

Given the inherent motion smoothness and semantic similarity between adjacent frames in continuous video, assume that frames with higher similarity to their neighboring frames are more representative. Hence, an important challenge is measuring the similarity between video frames. Based on the above discussion, the problem of finding representative keyframes can be formulated as finding the set of N frames with maximum similarity to their neighboring frames. Wei et al. [24], [25] observed that adversarial examples crafted on images can exhibit partial transferability to video action recognition models. This discovery implies a high-dimensional feature similarity between representations extracted by image and video models. Utilizing pretrained image models to extract features from individual frames is reasonable. The cosine similarity between frame-level features can then be compared as a proxy for representativeness. Then the optimization target in Equation 5 can be expressed as Equation 67:

$$\delta_{global} = [\delta_1, \delta_2, \dots, \delta_T] \quad (6)$$

$$\delta_i = \begin{cases} \delta_{key_i} & i \in \text{key frame list} \\ \delta_{key_j}, j = \underset{j}{\operatorname{argmin}}(|x_i, x_j|) & i \notin \text{key frame list} \end{cases} \quad (7)$$

Where δ_{key_i} denotes the key perturbation on i_{th} frame, $|x_i, x_j|$ denotes cosine similarity between frame i and j , the computational formula is given in Equation 8. $e(\cdot)$ denotes imagenet1k-pretrained image feature extractor.

$$|x_i, x_j| = \frac{e(x_i) \cdot e(x_j)}{|e(x_i)| |e(x_j)|} = \frac{\sum_{n=1} e(x_i)_n \times e(x_j)_n}{\sqrt{\sum_{n=1} (e(x_i)_n)^2} \times \sqrt{\sum_{n=1} (e(x_j)_n)^2}} \quad (8)$$

The keyframe selection algorithm is shown as Algorithm 1.

Given T input video frames ($T \geq N$), initialize each frame as its group, resulting in T groups, each containing a single frame. Next, iteratively compute the cosine similarity between every adjacent group, respectively, as shown in Equation 9. Then, the most similar pair of groups is merged into one group. This grouping and merging process is repeated, reducing the number of groups by one each iteration. The algorithm

Algorithm 1 keyframe select

Input: video frame list $X = \{x_1, x_2, \dots, x_T\}$, pretrained image feature extractor e , required keyframes number N

Output: keyframe list L

```

1: Frame groups  $G \leftarrow [[x_1], [x_2], \dots, [x_T]]$ 
2: while  $\text{len}(G) > N$  do
3:   for all  $g \in G$  do
4:     Calculate cosine similarity between  $g$  and  $g + 1$ 
5:   end for
6:   Find the biggest cosine similarity between  $g_i, g_j$ 
7:   Converge  $g_i, g_j$  to one group
8: end while
9: for all  $g \in G$  do
10:  for all  $x \in g$  do
11:    Calculate cosine similarity between  $x$  and other frames in  $g$ 
12:  end for
13:  Choose the frame  $x$  with biggest cosine similarity
14:   $L.append(x)$ 
15: end for
16: return  $L$ 
```

terminates when the number of groups equals the desired number of keyframes N for the target model.

$$\text{sim}(g_i, g_j) = \frac{\sum_{x \in g_i} \sum_{x' \in g_j} |x, x'|}{\text{len}(g_i) + \text{len}(g_j)} \quad (9)$$

Finally, the most representative frame is selected from each group as the one with maximum average similarity to other frames in that group via Equation 10. This results in a final set of N diverse keyframes capturing the predominant concepts throughout the video.

$$\underset{x_i \in g}{\operatorname{argmax}} \frac{1}{\text{len}(g)} \sum_{x_j \in g, x_j \neq x_i} |x_i, x_j| \quad (10)$$

2) *Attention-based Mask*: The model's attention region is calculated for selected keyframes, and attacks are only performed at the highest attention region. Given the following assumptions: due to the translation invariance of the convolutional neural network, after the feature extraction by multi-layer convolution, the position of attention regions on the feature maps should be the same as the images, which play more influence on the prediction result of target model [26] [27]. Furthermore, to make the adversarial attack more efficient, the region of the more significant gradient should be selected as the focus region so that small changes will significantly impact the final output.

For the input samples, the feature extraction is first performed using the feature extractor of the network, and the feature maps $x_{feature}$ are obtained. After that, the gradient of the feature map about the output confidence corresponding to the ground-truth label is calculated, as shown in Equation 11 where $f(x)_y$ denotes the confidence score of the label y in the model's output about x .

$$g = \nabla_{f(x)_y} (x_{feature}) \quad (11)$$

Finally, as shown in Equation 12, threshold filtering is performed, setting the mask's pixels' value to 1 when it is greater than the threshold. Otherwise, it is set to 0. The attack threshold th is a hyperparameter set by the attacker. For complex samples and models, th can be lowered to achieve a successful attack. Conversely, raising th enhances the imperceptibility of perturbations. The threshold provides a tunable control between attack transferability and perturbation concealment.

$$mask_{c,t,w,h} = \begin{cases} 1, & g_{norm_{c,t,w,h}} > th \\ 0, & g_{norm_{c,t,w,h}} \leq th \end{cases} \quad (12)$$

3) *Temporal Recurrent Translation Attack*: When diffusing the perturbations on keyframes, strategies are needed to enhance the generalization of adversarial perturbations since the semantic content on different frames must be considered. Considering that the main difference between different frames in the same video sample is the information in the temporal dimension, a temporal recurrent translation operation approach is used to enhance the transferability of adversarial perturbation during the attack on keyframes. In the adversarial attack against keyframes, the adversarial sample is panned in the time dimension once after each iteration cycle, as shown in Equation 13. Where $x_{adv_i}^m$ means the i_{th} frame of the adversarial keyframes generated in the m_{th} epoch.

$$Trans(x_{adv}^m) = \{x_{adv_N}^m, x_{adv_0}^m, x_{adv_1}^m, \dots, x_{adv_{N-1}}^m\} \quad (13)$$

Finally, the adversarial perturbation is updated for the panned adversarial sample keyframes, and the iterative method is given by Equation 14,15,16.

$$x_{adv}^{m+1} = Trans(x_{adv}^m) + mask \odot \alpha \cdot \text{sign}(\nabla_{Trans(x_{adv}^m)} J(Trans(f(x_{adv}^m)), y)) \quad (14)$$

$$x_{adv}^{m+1} = Clip(x_{adv}^{m+1}, 0, 1) \quad (15)$$

$$x_{adv}^0 = x_{key} + mask \odot \mu, \mu \sim U(-\epsilon/2, \epsilon/2) \quad (16)$$

Where α represents step size, $\text{sign}(\cdot)$ denotes symbolic functions, $\text{sign}(x) = \begin{cases} 1, & x > 0 \\ 0, & x \leq 0 \end{cases}$, $J(\cdot)$ is the loss function, in this paper, cross-entropy loss function is adopted, which is shown in Equation 17, μ is the initial random noise obeying uniform distribution.

$$J(x, y) = - \sum_{c=1}^C \log \frac{\exp x_c}{\sum_{i=1}^C \exp x_i} y_c \quad (17)$$

where x is the input, y is label, C is the number of classes, x_c is the confidence score corresponding to class c . The above temporal recurrent translation operation can greatly enhance the migration between frames in the same video adversarial perturbations. After generating adversarial perturbations on keyframes, replicate the perturbations to non-keyframes within the same group.

In summary, the attention-based sparse global attack method is shown by Algorithm 2.

Algorithm 2 Attention-based Sparse Global Attack

Input: video frame list $x = \{x_0, x_1, x_2, \dots, x_T\}$, video action recognition model f , required keyframes number N , true label y , attention threshold th , attack perturbation scale ϵ , maximum iteration period M , attack iteration step α

Output: global video adversarial sample x_{adv}

- 1: $x, x_{key} = S(x, N)$, select keyframe via algorithm 1
- 2: Get $mask$ by Equation 11, 12
- 3: **while** $epoch < M$ **do**
- 4: $x_{key_{adv}}^{epoch} \leftarrow \text{Attack}(x_{key_{adv}}^{epoch-1}, mask)$
- 5: **end while**
- 6: $x_{adv} \leftarrow \text{Diffusion}(x_{key_{adv}}^M)$, Diffusion perturbation to non-keyframes.
- 7: **return** x_{adv}

IV. EXPERIMENT & ANALYSIS

A. Experiment Setting

1) *Dataset*: Three public data sets are selected in this paper: UCF-101 [28], HMDB-51 [29] and Kinetics-400 [30]. UCF-101 is an action recognition data set of realistic action videos collected from YouTube, with 13320 videos from 101 action categories. HMDB-51 contains 6849 clips divided into 51 action categories, each containing a minimum of 101. Kinetics-400 contains 400 human action classes, with at least 400 video clips for each action.

2) *Video Recognition Model*: Three CNN-structure based video action recognition models, I3D [12], R(2+1)D [13], and SlowFast [14], two ViT based models, MViTv1-base [16] and MViTv2-small [17], and two RNN based models, RViT-base [31] and ML-LSTM [32], are used as target models. All models were trained based on the mmaction2 platform [33] with pytorch framework, and all achieved the accuracy claimed by the model proposers on the above dataset. Table II shows the test accuracy for each model.

3) *Attack Setting*: As a comparison experiment, FGSM [18], PGD [19], MIFGSM [34], BIM [35], and AdamFGSM [36] are selected as traditional attacks. These attack methods were initially used to attack static images. However, with some simple modifications, they can be used to generate video adversarial samples.

For video adversarial attacks, Sparse Attack [3] and MSANet Attack [21] are selected as comparisons. These two attacks are both sparse attacks. Table III shows the above adversarial attack parameters.

4) *Evaluation Metrics*:: The following metrics are used to evaluate the different attack methods.

Attack Success Rate (ASR) is the attack success rate when perturbing individual samples to induce misclassification.

Mean absolute Perturbation(MAP): the mean perturbation of each pixel throughout the video, which is calculated via Equation 18. Where $x_{(i,j,t)}$ denotes pixel value on position (i, j, t) of video.

$$MAP = \frac{\sum_{i,j,t} |x_{(i,j,t)} - x_{adv(i,j,t)}|}{T \times W \times H} \quad (18)$$

TABLE II: Test accuracy of the target models

Models	First N Frame (%)			Random Sample (%)			Random First Frame (%)		
	UCF-101	HMDB-51	Kinetics-400	UCF-101	HMDB-51	Kinetics-400	UCF-101	HMDB-51	Kinetics-400
I3D	94.64	70.13	74.80	92.06	69.61	73.95	89.78	69.10	74.12
R(2+1)D	95.56	74.90	75.46	95.04	72.68	74.33	93.43	70.40	75.09
SlowFast	95.77	75.62	75.55	95.55	74.49	75.48	92.96	73.84	74.93
MViTv1-base	—	—	78.48	—	—	76.15	—	—	75.88
MViTv2-small	—	—	80.76	—	—	77.96	—	—	76.06
RViT-base	—	—	77.65	—	—	77.93	—	—	77.58
ML-LSTM	—	—	72.46	—	—	71.98	—	—	70.63

TABLE III: Adversarial attack parameters

Attack Method	Parameters
FGSM [18]	Epsilon=16/255
PGD [19]	Epsilon=16/255, step size=1.6/255, iteration number=10
MIFGSM [34]	Epsilon=16/255, step size=1.6/255, iteration number=10, decay=1.0
BIM [35]	Epsilon=16/255, step size=1.6/255, iteration number=10
Sparse Attack [3]	Epsilon=16/255, step size=1.6/255, iteration number=10, sparsity=0.25
AdamFGSM [36]	Epsilon=16/255, step size=1.6/255, iteration number=10, decay=1.0
MSANet [21]	Epsilon=16/255, step size=1.6/255, iteration number=10
Ours	Epsilon=16/255, step size=1.6/255, iteration number=10

Global Attack Success Rate(GASR): the global attack success rate is calculated according to Equation 4, and in this experiment, the sampling number $k = 10$. As described in section 3.1, two random sampling methods will be used to evaluate the global attack performance of attack methods. For the "Random Sample" method, ten times random sampling will be carried out for each video, and the number of frames taken each time is N , and then, the N frames are combined into a new sample in chronological order for testing. Finally, the average attack success rate is calculated according to Equation 4. For each video, a frame from the first frame to the $(T-N)$ th frame is randomly selected as the starting frame, and then $N - 1$ frames are successively taken from this frame, and these N frames are combined into a new sample for testing. As shown in Table II, classification accuracy remains unchanged, mainly under original and random sampling strategies across datasets and models. This phenomenon indicates our proposed random global attack evaluation protocol does not inherently bias baseline accuracy.

B. Experiment Preparation

In this section, for UCF-101 and HMDB-51, attacks target CNN-based video action recognition models, such as I3D [12], R(2+1)D [13] and SlowFast [14], while ViT architectures are evaluated on Kinetics-400, such as MViTv1-base [16] and MViTv2-small [17]. Furthermore, we benchmark several classic image attack methods extended to videos, including FGSM [18], PGD [19], MIFGSM [34], and BIM [35] to provide baselines. In addition, recent sparse video attack techniques like Sparse Attack [3], Adam-FGSM [36], and MSANet [21] are compared to our proposed approach.

Since the attack, as mentioned above, methods were not initially designed for global video adversarial examples, only single samples are perturbed. The following strategies are adopted to make a fair comparison. A video of length T frames is divided into m non-overlapping clips of N frames each. The first clip is attacked traditionally, and the adversarial perturbation generated is copied over to the remaining $m - 1$ clips. The m distorted clips are recombined to form the full adversarial video. Video-level examples can be crafted without excessive computation by propagating the noise from a single attacked clip. For I3D, R(2+1)D, and SlowFast models, the length of the clip is 32; for MViTv1-base and MViTv2-small, the size of the clip is 16. Assume that a well-performing model will exhibit consistent classification accuracy for short video datasets used in this study under different sampling strategies. This phenomenon is based on the observation that frames in most dataset videos belong to the same action label. Our experiments validate this hypothesis, showing slight variation in accuracy when models are evaluated using different frame sampling approaches on a given test set. We attribute this to the assessed datasets' limited diversity and length of actions.

C. Attack Performance Comparison

As shown in Table IV, V and VI, both conventional image attacks and sparse video-specific attacks demonstrate limited global attack ability across datasets and models. Notably, their per-sample attack success is consistent with original reports, often over 90%. For example, Sparse Attack reaches over 99% ASR against I3D on the UCF-101 dataset. However, under Random Sample evaluation, the success rate drops to 20.56% and further to 14.15% with the Random First Frame strategy. This phenomenon holds across models and datasets.

It indicates the perturbations generated by Sparse Attack have limited intra-frame transferability. Despite propagating adversarial noise globally, attack success is not maintained. Our proposed evaluation protocol reveals such weaknesses invisible from single sample analysis. This result validates its necessity for benchmarking video attacks where the whole sequence can be sampled.

In contrast, our approach induces perturbations that transfer within the sequence, achieving high global attack success rates under rigorous random sampling conditions. For example, for the R(2+1)D model attack on the UCF-101 dataset, our attack method obtains state-of-the-art ASR and GASR. Furthermore, strong performance holds across dataset scales up to Kinetics-400.

Our approach demonstrates excellent stealthiness, competitive with dedicated sparse attack methods given matched perturbation budgets and iterations. Under the same thresholds, our perceptibility is slightly lower than the state-of-the-art MSANet, but on HMDB-51 against the I3D model, our method even performs better. This result proves the efficacy of our selective attention mechanism for imperceptible noise optimization.

TABLE IV: Experiments on UCF-101 Dataset

Models	Attack Methods	ASR	MAP	GASR	
				Random Sample	Random First Frame
I3D	FGSM [18]	78.65	0.0513	32.02	22.01
	PGD [19]	97.61	0.0328	23.37	17.50
	MIFGSM [34]	94.23	0.0489	33.93	22.30
	BIM [35]	93.11	0.0277	22.81	8.46
	Sparse Attack [3]	99.01	0.0124	20.56	14.15
	AdamFGSM [36]	99.47	0.0198	34.12	24.25
	MSANet [21]	98.04	0.0103	24.03	12.22
	Ours	96.21	0.0109	71.96	61.70
R(2+1)D	FGSM [18]	91.67	0.0543	59.88	35.89
	PGD [19]	97.78	0.0284	55.24	25.33
	MIFGSM [34]	96.49	0.0238	72.62	44.02
	BIM [35]	98.21	0.0268	56.67	32.24
	Sparse Attack [3]	91.09	0.0545	51.46	31.09
	AdamFGSM [36]	91.52	0.0367	62.18	36.35
	MSANet [21]	91.70	0.0099	36.32	15.42
	Ours	99.25	0.0139	78.65	50.06
SlowFast	FGSM [18]	71.76	0.0540	27.29	8.79
	PGD [19]	96.55	0.0283	32.24	12.80
	MIFGSM [34]	90.22	0.0233	42.24	25.07
	BIM [35]	91.26	0.0304	33.53	17.52
	Sparse Attack [3]	94.48	0.0547	41.66	26.49
	AdamFGSM [36]	91.70	0.0201	34.18	17.70
	MSANet [21]	94.30	0.0101	19.21	10.33
	Ours	90.45	0.0118	68.68	51.32

D. Adversarial Transferability Evaluation

In the tests targeting black-box attacks, we considered two scenarios. One is the cross-model transferability test, where adversarial examples are generated on model A and their attack

TABLE V: Experiments on HMDB-51 Dataset

Models	Attack Methods	ASR	MAP	GASR	
				Random Sample	Random First Frame
I3D	FGSM [18]	95.83	0.0560	47.08	41.75
	PGD [19]	98.96	0.0298	35.56	29.16
	MIFGSM [34]	98.02	0.0233	48.19	43.07
	BIM [35]	99.99	0.0195	37.50	32.49
	Sparse Attack [3]	97.73	0.0566	49.56	43.12
	AdamFGSM [36]	98.08	0.0359	64.21	57.45
	MSANet [21]	99.05	0.0105	28.44	29.00
	Ours	99.31	0.0098	92.00	88.89
R(2+1)D	FGSM [18]	95.12	0.0589	65.61	57.31
	PGD [19]	96.44	0.0300	76.34	67.00
	MIFGSM [34]	96.00	0.0250	86.59	76.85
	BIM [35]	98.98	0.0196	77.20	67.26
	Sparse Attack [3]	96.09	0.0600	78.65	72.59
	AdamFGSM [36]	99.48	0.0298	70.74	65.03
	MSANet [21]	99.19	0.0103	67.08	49.17
	Ours	95.62	0.0119	95.21	80.62
SlowFast	FGSM [18]	90.59	0.0589	62.12	52.64
	PGD [19]	98.58	0.0300	46.71	41.01
	MIFGSM [34]	95.76	0.0247	63.41	56.85
	BIM [35]	95.01	0.0195	48.24	41.27
	Sparse Attack [3]	95.20	0.0222	48.57	41.27
	AdamFGSM [36]	95.69	0.0236	65.12	58.31
	MSANet [21]	97.36	0.0104	39.44	32.67
	Ours	99.80	0.0113	93.00	82.67

TABLE VI: Experiments on Kinetics-400 Dataset

Models	Attack Methods	ASR	MAP	GASR	
				Random Sample	Random First Frame
MVITv1-base	FGSM [18]	71.49	0.0447	25.67	8.53
	PGD [19]	83.39	0.0260	31.65	12.14
	MIFGSM [34]	86.56	0.0225	37.30	24.49
	BIM [35]	83.56	0.0267	33.21	16.53
	Sparse Attack [3]	90.00	0.0484	34.23	25.46
	AdamFGSM [36]	83.72	0.0199	27.94	15.39
	MSANet [21]	87.15	0.0092	18.71	9.20
	Ours	87.76	0.0098	61.63	56.26
MVITv2-small	FGSM [18]	79.07	0.0502	24.63	7.59
	PGD [19]	87.71	0.0250	26.92	10.90
	MIFGSM [34]	85.79	0.0215	38.41	21.18
	BIM [35]	86.82	0.0236	26.18	15.87
	Sparse Attack [3]	84.89	0.0488	34.19	20.98
	AdamFGSM [36]	86.18	0.0168	27.61	16.64
	MSANet [21]	86.08	0.0082	17.51	7.93
	Ours	83.80	0.0103	65.41	57.08

success rate is tested on model B. The other is the cross-dataset transferability test, in which adversarial examples are generated on a model trained on dataset A, and then their attack success rate is tested on a model trained on dataset B.

TABLE VII: Cross-model adversarial transferability evaluation

Source Models	Attack Methods	Target Models			
		R(2+1)D	SlowFast	MViTv2-tiny	ML-LSTM
I3D	Sparse Attack [3]	38.90	43.82	11.49	15.66
	MSANet [21]	39.31	30.59	13.03	23.69
	ASGA (Ours)	53.23	51.78	26.11	40.85
MViTv1-base	Sparse Attack [3]	25.11	28.49	40.05	30.18
	MSANet [21]	27.00	27.24	57.47	26.76
	ASGA (Ours)	27.33	31.63	55.40	38.81
RViT-base	Sparse Attack [3]	23.20	22.14	56.35	25.07
	MSANet [21]	23.36	26.89	56.79	29.96
	ASGA (Ours)	19.52	19.41	58.91	38.70

1) *Cross Model Transferability*: For the cross-model transferability test, we selected Kinetics-400 as both the training and testing dataset, and used three models, namely I3D, MViTv1-base, and RViT-base, as the source models. Subsequently, attack tests were conducted on target models with different architectures. The experimental data are presented in Table VII. For comparison, we also selected two video adversarial attack methods, namely Sparse Attack and MSANet, to conduct the transferability test for black-box attacks.

From the experimental results, the ASGA method proposed in this paper exhibits overall better transferability than the two attack methods, Sparse Attack and MSANet. When using the I3D model as the source model, it achieves the best test results across all different target models. When taking MViTv1-base as the source model, its performance in the test on MViTv2-tiny is slightly inferior to that of MSANet. When the RViT-base serves as the source model, it shows relatively poor performance in the R(2+1)D and SlowFast models.

It can be clearly observed from the experimental data that the level of transferability is closely related to the structural similarity between the source model and the target model. For example, adversarial examples generated by the I3D model, which is based on a CNN structure, exhibit the best transferability in the R(2+1)D and SlowFast models that also adopt a CNN-based structure. However, their attack success rates are relatively low in the MViTv2-tiny model with a ViT structure and the ML-LSTM model with an RNN structure. This trend is equally evident in ViT models and RNN models, and it is consistent across adversarial examples generated by different attack methods.

2) *Cross Dataset Transferability*: To validate the transferability of ASGA attacks across different video datasets, we used UCF-101 as training dataset, while HMDB-51 and Kinetics-400 were employed as target dataset. All models used for adversarial generation are trained on UCF-101. I3D and MViTv1-base models with identical architectures were used as target models for adversarial example generation and testing, thereby eliminating the influence of model architecture on cross-dataset adversarial attack transferability testing. To mitigate biases arising from label inconsistency, 13 semantically consistent action labels across HMDB-51, Kinetics-400, and UCF-101 were selected for testing, as detailed in Table VIII.

As comparative attack methods for cross-dataset transfer testing, we selected two video attack methods, Sparse Attack and MSANet, for comparison, with attack parameters set in

TABLE VIII: Used labels in cross-datasets adversarial transferability evaluation.

No.	UCF-101 Label	HMDB-51 Label	Kinetics-400 Label
1	Fencing	fencing	sword fighting
2	Push Ups	push up	push up
3	Pull Ups	pull up	pull ups
4	Diving	dive	diving cliff
5	Rock Climbing Indoor	climb	rock climbing
6	Throw Discus	throw	throwing discus
7	Soccer Juggling	kick ball	kicking soccer ball
8	Walking with a dog	walk	walking the dog
9	Frisbee Catch	catch	catching or throwing frisbee
10	Biking	ride bike	riding a bike
11	Horse Riding	ride horse	riding or walking with horse
12	Archery	shoot bow	archery
13	Boxing-Punching Bag	punch	punching bag

accordance with the original papers. For each label category, 100 samples that could be correctly identified by both the model trained on UCF-101 and the model trained on HMDB-51 were selected, totaling 1,300 video clips as the test dataset.

TABLE IX: Cross-datasets adversarial transferability evaluation

Target Datasets	Target Models	Attack Methods	ASR	GASR	
				Random Sample	Random First Frame
HMDB-51	I3D	Sparse Attack	78.54	31.23	29.08
		MSANet	73.15	28.15	35.00
		ASGA (Ours)	85.31	79.08	79.54
Kinetics-400	MViTv1-base	Sparse Attack	44.00	11.62	6.54
		MSANet	45.31	12.46	6.92
		ASGA (Ours)	66.85	57.00	48.31

Table IX presents the results of cross-dataset adversarial transferability tests for video adversarial attacks. A significant reduction in attack success rates is observed when transferring attacks from source-trained models to target-trained ones, highlighting the challenges inherent in cross-dataset adversarial attacks. This phenomenon is primarily attributed to distribution disparities between training sets of different datasets, which persist despite semantically consistent action labels. Such disparities reduce the effectiveness of source-generated adversarial examples when applied to target-trained models.

Despite these challenges, the ASGA attack method proposed in this study demonstrates superior performance. In attacks against HMDB-51, ASGA achieved global attack success rates of 0.7908 and 0.7954 under the Random Sample and Random First Frame scenarios, respectively. These results significantly outperform those of Sparse Attack (0.3123 and 0.2908) and MSANet (0.3500 and 0.2815).

This superiority is attributed to two key mechanisms in ASGA: 1) prioritization of critical region selection during adversarial noise generation, which concentrates noise in model-attended areas that exhibit consistent discriminative patterns across datasets; and 2) temporal dimension translation operations that reduce sensitivity to temporal inconsistencies, thereby enhancing noise transferability. Together, these

mechanisms endow ASGA with robust cross-dataset transfer capabilities.

In summary, the ASGA method demonstrates favorable performance in cross-model and cross-dataset video attacks. However, adversarial transferability—particularly for global attacks—remains a significant challenge, necessitating further research.

E. Long Video Attack

Notable dataset-specific trends emerge in global attack performance despite comparable per-sample success rates. For instance, our method applied to the R(2+1)D model achieves an Attack Success Rate (ASR) exceeding 97% on both HMDB-51 and UCF-101, yet the global attack success rate is over 15% higher on HMDB-51. We hypothesize this discrepancy arises from differing average video lengths. As illustrated in Figure 2, HMDB-51 exhibits shorter median clip durations compared to other datasets. A consistent inverse relationship is observed between global attack success and sequence length: datasets with longer videos yield lower global success rates. This phenomenon is attributed to reduced intra-frame transferability across distant keyframes. Specifically, larger frame groups accumulate greater inter-frame variability, diminishing the representativeness of keyframes and thereby weakening the transferability of adversarial perturbations optimized on these frames to the entire sequence.

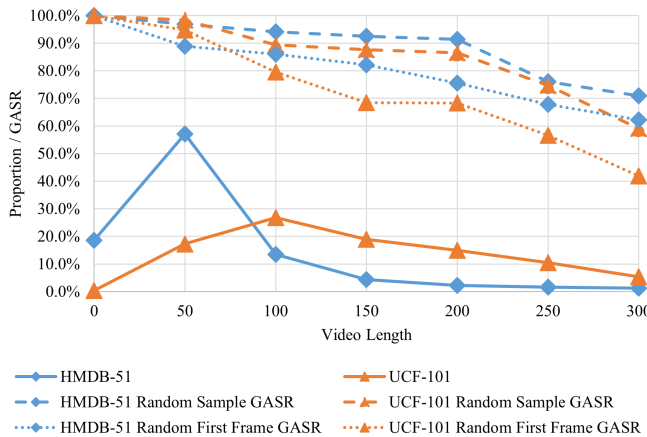


Fig. 2: Global attack success rate trends for our method against R(2+1)D as video length varies on UCF-101 and HMDB-51

A simple yet effective approach to maintaining high global attack rates on long videos is segmenting them into shorter segments and applying our method separately to each. The perturbed segments can then be concatenated to construct the entire adversarial sequence. Despite slightly increased computation, this remains an efficient attack strategy. As Figure 2 shows, segment lengths around 200 frames still achieve high success rates. We validate this approach by attacking videos over 500 frames in UCF-101, with results in Table X. Segmenting the long videos into lengths of 200 frames sustains global success over 80%, significantly improving upon direct application to the complete sequence.

TABLE X: Experiments on long videos (> 500 frames)

Models	Attack Methods	ASR	MAP	GASR	
				Random Sample	Random First Frame
I3D	Sparse Attack [3]	96.82	0.0312	34.70	38.74
	MSANet [21]	95.34	0.0147	33.90	30.24
	Ours	98.92	0.0142	89.60	80.04
R(2+1)D	Sparse Attack [3]	99.02	0.0340	33.44	36.64
	MSANet [21]	99.56	0.0117	38.60	37.58
	Ours	95.20	0.0141	89.40	76.16
SlowFast	Sparse Attack [3]	99.12	0.0327	37.42	32.64
	MSANet [21]	97.59	0.0109	32.18	31.64
	Ours	98.57	0.0113	84.16	78.44

TABLE XI: Adversarial example robustness evaluation on Kinetics-400 dataset

Defense Methods	I3D			MViTv1-Base		
	ASR	GASR		ASR	GASR	
		Random Sample	Random First Frame		Random Sample	Random First Frame
None	95.48	77.42	67.90	83.09	59.75	54.03
JPEG (qp=90)	93.67	74.99	66.42	79.97	59.12	51.32
JPEG (qp=70)	91.57	73.52	60.32	71.93	50.89	47.98
JPEG (qp=50)	89.16	73.26	59.59	70.09	48.58	45.65
Gaussian	93.61	73.99	61.47	77.54	53.02	45.38
Super	90.30	69.20	64.29	77.84	55.94	45.50
Random	91.10	68.24	65.17	70.97	54.08	49.95
HEVC-AD	71.44	52.94	58.18	77.25	53.21	49.82
NIP	67.33	34.47	38.27	77.28	41.11	32.96
Average	87.07	66.45	60.18	76.22	52.86	46.95

F. Adversarial Example Robustness Evaluation

In implementing adversarial example techniques, the robustness of adversarial examples is a critical consideration. We investigated the robustness against both adversarial example defense methods and conventional image processing techniques.

We selected typical image processing algorithms—Random Resize and Padding (Random) [37], Gaussian Blurring (Gaussian) [38], JPEG Compression (JPEG) [39], and Super-Resolution (Super) [40]—as well as two state-of-the-art adversarial example detection and defense methods: HEVC-AD [41] and NIP [42], with parameters consistent with their original papers.

The robustness test results of the proposed ASGA method are shown in Table XI, conducted on the Kinetics-400 dataset using the I3D and MViTv1-base models, respectively. "None" indicates no adversarial defense is applied.

Existing adversarial example defense methods demonstrate varying degrees of efficacy against ASGA-generated attacks. The NIP paper's adversarial example detection method yields the most pronounced defense: on the I3D model, ASGA's attack performance and global success rate are reduced by up to 28.15% and 42.95%, respectively, highlighting the impact of defense strategies.

In the MViTv1-base model, the NIP and HEVC-AD methods exhibit diminished effectiveness. HEVC-AD's joint

compression- and pixel-domain feature detection struggles against ViT architectures, as adversarial noise forms blocky patterns that undermine encoding-based detection.

NIP's parameter-importance analysis similarly falters with ViT models, as their high parameter count limits statistical feature efficacy in capturing neuron sensitivity.

Collectively, these results underscore ASGA's vulnerability to existing defenses, reflected in its low global success rate. This confirms the enduring challenge of global video adversarial attacks, which we aim to address through targeted enhancements in future research.

G. Random Sample Strategy

In Table IV, V, VI and X, a clear trend emerges that random sample evaluation achieves markedly higher attack success rates compared to the random first frame. The keyframe selection strategy causes this. Let $\delta_{key} = [\delta_{key_1}, \delta_{key_2}, \dots, \delta_{key_N}]$ denotes the adversarial perturbation add to the keyframe sets x_{key} . For simplicity, assume the action in a video is continuous and uniform. When grouping frames by feature similarity, the expected group length is T/N where T is the video length, and N is the number of groups. Since one keyframe is selected per group, the expected distance between keyframes is also T/N . Let $x_{adv} = [x_1 + \delta_{key_1}, x_2 + \delta_{key_1}, \dots, x_{(i-1) \times T/N + 1} + \delta_{key_i}, x_{(i-1) \times T/N + 2} + \delta_{key_i}, \dots]$ denotes the global adversarial example generated using these keyframes. The global attack success rate is higher when the randomly sampled clip contains more distinct key adversarial frames used to construct the perturbation. Conversely, the success rate decreases when fewer keyframes are present.

Given the earlier assumption of uniform action in the video, the variance of the keyframe indices sampled under random sample evaluation can be derived as Equation 19:

$$\begin{aligned} E_{RS}(\delta_{key}) &= \sqrt{\frac{1}{N} \sum_{j=1}^N (j - \frac{N+1}{2})^2} \\ &= \sqrt{\frac{(N+1)(N+2)}{24}} \end{aligned} \quad (19)$$

the variance of the keyframe indices sampled under random first frame evaluation can be derived as Equation 20:

$$\begin{aligned} E_{RFF}(\delta_{key}) &\leq \sqrt{\frac{1}{N} \sum_{i=0}^{\lceil \frac{N^2}{T} \rceil} \frac{T}{N} \left(i - \frac{T}{2N} \left(\frac{N^2}{T} + 1 \right) \right)^2} \\ &= \sqrt{\frac{T}{N^2} \sum_{i=0}^{\lceil \frac{N^2}{T} \rceil} \left(i - \frac{N}{2} - \frac{T}{2N} \right)^2} \\ &\approx \sqrt{\frac{T}{N^2} \cdot \frac{N^4}{12T}} \\ &= \sqrt{\frac{N^2}{12} \cdot \frac{T}{N}} \\ &= \sqrt{\frac{T}{8}} \end{aligned} \quad (20)$$

The complete proof and derivations leading to the previously stated variance formula are provided in the Appendix.

According to the Settings in the experiment, for CNN-based models, $N = 32$, $E_{RS} \approx 6.84$, when $T < 374$, $E_{RFF} \leq E_{RS}$, which makes the global attack success rate in random first frame condition is lower. In the same way, for ViT based models, $N = 16$, $E_{RS} \approx 3.57$, when $T < 102$, $E_{RFF} \leq E_{RS}$.

The theoretical sampling analysis also explains the empirical observation of greater divergence in attack success rates between the random sample and random first frame evaluation as video length increases across datasets. As video sequences grow longer, each group contains more frames, which leads to declining global attack success through two key factors: 1. Weakened frame representativeness. The larger the group, the less each frame encapsulates the global context. Optimization of non-representative frames hurts generalization. 2. Reduced adversarial subsample diversity. Given an initial random start frame, there are fewer possible keyframe combinations to sample within a group, and this operation limits diversity in perturbations seen during optimization. Together, these effects result in lower global attack success on longer videos. Unrepresentative examples need to capture the breadth of features required for transferability. And limited diversity in adversarial perspectives causes overfitting.

H. Ablation Studies

This section investigates the impact of key components in our proposed attack method—specifically keyframe selection, attention masking, and recurrent temporal translation—on overall performance. By isolating these factors and evaluating their contributions, we aim to clarify the underlying mechanisms driving attack success, enabling targeted improvements. All experimental settings, except for the components under analysis, remain consistent with previous experiments. Ablation studies are performed on the UCF-101 dataset targeting the I3D model, with results presented in Table XII.

1) *Keyframe Selection*: We compare the feature similarity-based keyframe selection algorithm with uniform sampling. The feature similarity-based approach achieves significantly lower perturbation magnitudes with only a negligible reduction in attack success rate. This is because it prioritizes frames that better capture critical temporal information, whereas uniform sampling may include redundant frames, leading to unnecessary perturbations.

2) *Attention Masking*: Compared to the absence of masking, attention masking yields substantially lower perturbation magnitudes with minimal sacrifice in attack success rate. Attention regions target input areas most sensitive to attack optimization; higher sensitivity implies greater influence on model outputs. Without masking, even insensitive regions accumulate excessive, unnecessary perturbations that become hard to reduce during optimization. By focusing perturbations on relevant areas, masking enhances imperceptibility while maintaining effective attack performance, balancing stealth and potency.

3) *Recurrent Temporal Translation*: Recurrent temporal translation is another critical component: frame shifting enforces perturbation effectiveness across varying temporal arrangements, improving generalization. It distributes gradients

TABLE XII: Ablation studies experiment

keyframe Selection	Attention Mask	Recurrent Temporal Translation	MAP	GASR	
				Random Sample	Random First Frame
Feature Similarity Select	✓	✓	0.0109	71.96	61.70
Uniform Select	✓	✓	0.0113	50.62	38.87
Feature Similarity Select	✗	✓	0.0387	73.44	65.70
Feature Similarity Select	✓	✗	0.0107	32.77	17.45

across keyframes, mitigating inter-frame discrepancies and enhancing the transferability of shared perturbations. Notably, video recognition models exhibit low sensitivity to temporal order—even full frame scrambling causes minimal accuracy degradation. Attack performance degraded most significantly without translation compared to other factors, highlighting its role in enabling within-sample transferability. Temporal shifts induce invariance to sequence permutations, making perturbations robust to compositional changes by averaging gradient information over permutations and avoiding overfitting. In summary, temporal translation boosts attack success by enhancing perturbation resilience to varying frame combinations through simple data augmentation.

V. CONCLUSION

Existing video attack methods overlook the impact of sampling methods on adversarial transferability. Perturbations fail to maintain effectiveness when models alter frame selection. Our proposed method addresses this problem through a global attack approach using similarity-based keyframe selection and temporal translations during optimization. Attention masking further improves imperceptibility. Experiments across datasets and target models demonstrate greater robustness compared to prior arts. This work introduces the global attack success rate (GASR) to measure video attack performance from this new perspective.

REFERENCES

- [1] N. Afaq, N. Akhtar, W. Liu, M. Shah, and A. Mian, “Language model agnostic gray-box adversarial attack on image captioning,” *IEEE Trans. Inf. Forensics Secur.*, vol. 18, pp. 626–638, 2023.
- [2] Y. Li, S. Ruan, H. Qin, S. Deng, and M. A. El-Yacoubi, “Transformer based defense GAN against palm-vein adversarial attacks,” *IEEE Trans. Inf. Forensics Secur.*, vol. 18, pp. 1509–1523, 2023.
- [3] X. Wei, J. Zhu, S. Yuan, and H. Su, “Sparse adversarial perturbations for videos,” in *The Thirty-Third AAAI Conference on Artificial Intelligence, AAAI 2019, The Thirty-First Innovative Applications of Artificial Intelligence Conference, IAAI 2019, The Ninth AAAI Symposium on Educational Advances in Artificial Intelligence, EAAI 2019, Honolulu, Hawaii, USA, January 27 - February 1, 2019*. AAAI Press, 2019, pp. 8973–8980.
- [4] R. Mu, L. S. Marcolino, Q. Ni, and W. Ruan, “Enhancing robustness in video recognition models: Sparse adversarial attacks and beyond,” *Neural Networks*, vol. 171, pp. 127–143, 2024.
- [5] W. Deng, C. Yang, K. Huang, Y. Liu, W. Gui, and J. Luo, “Sparse adversarial video attack based on dual-branch neural network on industrial artificial intelligence of things,” *IEEE Transactions on Industrial Informatics*, vol. 20, no. 7, pp. 9385–9392, 2024.
- [6] G. Wu, J. Li, Y. Xu, Z. Shi, and X. Liu, “Video motion blur attack via grad-weighted and discrete-fusion based perturbation generation,” *IEEE Trans. Multim.*, vol. 27, pp. 3856–3868, 2025.
- [7] R. Pony, I. Naeh, and S. Mannor, “Over-the-air adversarial flickering attacks against video recognition networks,” in *IEEE Conference on Computer Vision and Pattern Recognition, CVPR 2021, virtual, June 19-25, 2021*. Computer Vision Foundation / IEEE, 2021, pp. 515–524.
- [8] Z. Zhao, K. Xu, X. Jiang, and T. Sun, “Research on video adversarial attack with long living cycle,” in *Uncertainty in Artificial Intelligence, Proceedings of the Thirty-Eighth Conference on Uncertainty in Artificial Intelligence, UAI 2022, ser. Proceedings of Machine Learning Research*, J. Cussens and K. Zhang, Eds., vol. 180. PMLR, 2022, pp. 2374–2382.
- [9] Y. Wang, M. Liu, Y. Ren, X. Zhang, and G. Feng, “Traffic sign attack via pinpoint region probability estimation network,” *Pattern Recognit.*, vol. 146, p. 110035, 2024.
- [10] Y. Wang, Y. Wang, and G. Feng, “Sample selection of adversarial attacks against traffic signs,” *Neural Networks*, vol. 180, p. 106698, 2024.
- [11] J. Li, Z. He, A. Luo, J. Hu, Z. J. Wang, and X. Kang, “Advad: Exploring non-parametric diffusion for imperceptible adversarial attacks,” in *Advances in Neural Information Processing Systems 38: Annual Conference on Neural Information Processing Systems 2024, NeurIPS 2024, Vancouver, BC, Canada, December 10 - 15, 2024*, A. Globersons, L. Mackey, D. Belgrave, A. Fan, U. Paquet, J. M. Tomczak, and C. Zhang, Eds., 2024.
- [12] J. Carreira and A. Zisserman, “Quo vadis, action recognition? A new model and the kinetics dataset,” in *2017 IEEE Conference on Computer Vision and Pattern Recognition, CVPR 2017, Honolulu, HI, USA, July 21-26, 2017*. IEEE Computer Society, 2017, pp. 4724–4733.
- [13] D. Tran, H. Wang, L. Torresani, J. Ray, Y. LeCun, and M. Paluri, “A closer look at spatiotemporal convolutions for action recognition,” in *2018 IEEE Conference on Computer Vision and Pattern Recognition, CVPR 2018, Salt Lake City, UT, USA, June 18-22, 2018*. Computer Vision Foundation / IEEE Computer Society, 2018, pp. 6450–6459.
- [14] C. Feichtenhofer, H. Fan, J. Malik, and K. He, “Slowfast networks for video recognition,” in *2019 IEEE/CVF International Conference on Computer Vision, ICCV 2019, Seoul, Korea (South), October 27 - November 2, 2019*. IEEE, 2019, pp. 6201–6210.
- [15] A. Vaswani, N. Shazeer, N. Parmar, J. Uszkoreit, L. Jones, A. N. Gomez, L. u. Kaiser, and I. Polosukhin, “Attention is all you need,” in *Advances in Neural Information Processing Systems*, I. Guyon, U. V. Luxburg, S. Bengio, H. Wallach, R. Fergus, S. Vishwanathan, and R. Garnett, Eds., vol. 30. Curran Associates, Inc., 2017.
- [16] H. Fan, B. Xiong, K. Mangalam, Y. Li, Z. Yan, J. Malik, and C. Feichtenhofer, “Multiscale vision transformers,” in *2021 IEEE/CVF International Conference on Computer Vision, ICCV 2021, Montreal, QC, Canada, October 10-17, 2021*. IEEE, 2021, pp. 6804–6815.
- [17] Y. Li, C. Wu, H. Fan, K. Mangalam, B. Xiong, J. Malik, and C. Feichtenhofer, “Mvitv2: Improved multiscale vision transformers for classification and detection,” in *IEEE/CVF Conference on Computer Vision and Pattern Recognition, CVPR 2022, New Orleans, LA, USA, June 18-24, 2022*. IEEE, 2022, pp. 4794–4804.
- [18] I. J. Goodfellow, J. Shlens, and C. Szegedy, “Explaining and harnessing adversarial examples,” in *3rd International Conference on Learning Representations, ICLR 2015, San Diego, CA, USA, May 7-9, 2015, Conference Track Proceedings*, Y. Bengio and Y. LeCun, Eds., 2015.
- [19] A. Madry, A. Makelov, L. Schmidt, D. Tsipras, and A. Vladu, “Towards deep learning models resistant to adversarial attacks,” in *6th International Conference on Learning Representations, ICLR 2018, Vancouver, BC, Canada, April 30 - May 3, 2018, Conference Track Proceedings*. OpenReview.net, 2018.
- [20] N. Carlini and D. A. Wagner, “Towards evaluating the robustness of neural networks,” in *2017 IEEE Symposium on Security and Privacy, SP 2017*. IEEE Computer Society, 2017, pp. 39–57.
- [21] G. Wu, Y. Xu, J. Li, Z. Shi, and X. Liu, “Imperceptible adversarial attack with multi-granular spatio-temporal attention for video action recognition,” *IEEE Internet of Things Journal*, pp. 1–1, 2023.

- [22] S. Li, A. Aich, S. Zhu, M. S. Asif, C. Song, A. K. Roy-Chowdhury, and S. V. Krishnamurthy, "Adversarial attacks on black box video classifiers: Leveraging the power of geometric transformations," in *Advances in Neural Information Processing Systems 34: Annual Conference on Neural Information Processing Systems 2021, NeurIPS 2021, December 6-14, 2021, virtual*, M. Ranzato, A. Beygelzimer, Y. N. Dauphin, P. Liang, and J. W. Vaughan, Eds., 2021, pp. 2085–2096.
- [23] Z. Wei, J. Chen, Z. Wu, and Y. Jiang, "Boosting the transferability of video adversarial examples via temporal translation," in *Thirty-Sixth AAAI Conference on Artificial Intelligence, AAAI 2022, Thirty-Fourth Conference on Innovative Applications of Artificial Intelligence, IAAI 2022, The Twelfth Symposium on Educational Advances in Artificial Intelligence, EAAI 2022 Virtual Event, February 22 - March 1, 2022*. AAAI Press, 2022, pp. 2659–2667.
- [24] —, "Cross-modal transferable adversarial attacks from images to videos," in *IEEE/CVF Conference on Computer Vision and Pattern Recognition, CVPR 2022, New Orleans, LA, USA, June 18-24, 2022*. IEEE, 2022, pp. 15 044–15 053.
- [25] Z. Wei, J. Chen, H. Zhang, L. Jiang, and Y. Jiang, "Adaptive temporal grouping for black-box adversarial attacks on videos," in *ICMR '22: International Conference on Multimedia Retrieval, Newark, NJ, USA, June 27 - 30, 2022*, V. Oria, M. L. Sapino, S. Satoh, B. Kerhervé, W. Cheng, I. Ide, and V. K. Singh, Eds. ACM, 2022, pp. 587–593.
- [26] S. Chen, Z. He, C. Sun, J. Yang, and X. Huang, "Universal adversarial attack on attention and the resulting dataset damagenet," *IEEE Trans. Pattern Anal. Mach. Intell.*, vol. 44, no. 4, pp. 2188–2197, 2022.
- [27] J. Zhu, F. Dai, L. Yu, H. Xie, L. Wang, B. Wu, and Y. Zhang, "Attention-guided transformation-invariant attack for black-box adversarial examples," *Int. J. Intell. Syst.*, vol. 37, no. 5, pp. 3142–3165, 2022.
- [28] K. Soomro, A. R. Zamir, and M. Shah, "UCF101: A dataset of 101 human actions classes from videos in the wild," *CoRR*, vol. abs/1212.0402, 2012.
- [29] H. Kuehne, H. Jhuang, E. Garrote, T. A. Poggio, and T. Serre, "HMDB: A large video database for human motion recognition," in *IEEE International Conference on Computer Vision, ICCV 2011, Barcelona, Spain, November 6-13, 2011*, D. N. Metaxas, L. Quan, A. Sanfeliu, and L. V. Gool, Eds. IEEE Computer Society, 2011, pp. 2556–2563.
- [30] W. Kay, J. Carreira, K. Simonyan, B. Zhang, C. Hillier, S. Vijayanarasimhan, F. Viola, T. Green, T. Back, P. Natsev, M. Suleyman, and A. Zisserman, "The kinetics human action video dataset," *CoRR*, vol. abs/1705.06950, 2017.
- [31] J. Yang, X. Dong, L. Liu, C. Zhang, J. Shen, and D. Yu, "Recurring the transformer for video action recognition," in *IEEE/CVF Conference on Computer Vision and Pattern Recognition, CVPR 2022, New Orleans, LA, USA, June 18-24, 2022*. IEEE, 2022, pp. 14 043–14 053.
- [32] A. Ullah, K. Muhammad, J. Del Ser, S. W. Baik, and V. H. C. de Albuquerque, "Activity recognition using temporal optical flow convolutional features and multilayer lstm," *IEEE Transactions on Industrial Electronics*, vol. 66, no. 12, pp. 9692–9702, 2019.
- [33] M. Contributors, "Openmmlab's next generation video understanding toolbox and benchmark," 2020.
- [34] Y. Dong, F. Liao, T. Pang, H. Su, J. Zhu, X. Hu, and J. Li, "Boosting adversarial attacks with momentum," in *2018 IEEE Conference on Computer Vision and Pattern Recognition, CVPR 2018, Salt Lake City, UT, USA, June 18-22, 2018*. Computer Vision Foundation / IEEE Computer Society, 2018, pp. 9185–9193.
- [35] A. Kurakin, I. J. Goodfellow, and S. Bengio, "Adversarial examples in the physical world," in *5th International Conference on Learning Representations, ICLR 2017, Toulon, France, April 24-26, 2017, Workshop Track Proceedings*. OpenReview.net, 2017.
- [36] J. Zhang, W. Qian, R. Nie, J. Cao, and D. Xu, "Generate adversarial examples by adaptive moment iterative fast gradient sign method," *Appl. Intell.*, vol. 53, no. 1, pp. 1101–1114, 2023.
- [37] C. Xie, J. Wang, Z. Zhang, Z. Ren, and A. L. Yuille, "Mitigating adversarial effects through randomization," in *6th International Conference on Learning Representations, ICLR 2018, Vancouver, BC, Canada, April 30 - May 3, 2018, Conference Track Proceedings*. OpenReview.net, 2018.
- [38] H. Brama and T. Grinshpoun, "Heat and blur: An effective and fast defense against adversarial examples," *CoRR*, vol. abs/2003.07573, 2020.
- [39] N. Das, M. Shanbhogue, S. Chen, F. Hohman, L. Chen, M. E. Kounavis, and D. H. Chau, "Keeping the bad guys out: Protecting and vaccinating deep learning with JPEG compression," *CoRR*, vol. abs/1705.02900, 2017.
- [40] A. Mustafa, S. H. Khan, M. Hayat, J. Shen, and L. Shao, "Image super-resolution as a defense against adversarial attacks," *IEEE Transactions on Image Processing*, vol. 29, pp. 1711–1724, 2020.
- [41] Z. Zhao, Y. Wang, K. Xu, T. Sun, and X. Jiang, "Hevc video adversarial samples detection via joint features of compression and pixel domains," *IEEE Transactions on Information Forensics and Security*, vol. 20, pp. 488–503, 2025.
- [42] R. Chen, H. Jin, H. Zheng, J. Chen, and Z. Liu, "Fight perturbations with perturbations: Defending adversarial attacks via neuron influence," *IEEE Transactions on Dependable and Secure Computing*, vol. 22, no. 2, pp. 1582–1595, 2025.



Zeyu Zhao received the B.S. degree in information security from Shanghai Jiaotong University, Shanghai, China, in 2020. He is currently pursuing the Ph.D. degree with the School of Electronic Information and Electrical Engineering, Shanghai Jiao Tong University, Shanghai, China. His research interests include adversarial examples and adversarial privacy protection.



Ke Xu received the Ph.D. degree in cyber space security from Shanghai Jiao Tong University, Shanghai, China, in 2019. He is currently an associate professor with the Institute of Cyber Space Security, the School of Electronic Information and Electrical Engineering, Shanghai Jiao Tong University, Shanghai. His research interests include action recognition, gait recognition and abnormal events detection. Dr. Xu is an IEEE member.



Tanfeng Sun received the Ph.D. degree in information and communication system from Jilin University, Changchun, China, in 2003. He is currently a Professor with the School of Electronic Information and Electrical Engineering, Shanghai Jiao Tong University, Shanghai, China. He had cooperated with Prof. Y.Q. Shi in New Jersey Institute of Technology, U.S.A, as a visiting scholar from Jul. 2012 to Dec. 2013. His research interests include Digital Forensics on Video Forgery, Digital Video Steganography and Steganalysis, Watermarking, and so on.



Xinghao Jiang received the Ph.D. degree in electronic science and technology from Zhejiang University, Hangzhou, China, in 2003. He is currently a Professor with the School of Electronic Information and Electrical Engineering, Shanghai Jiao Tong University, Shanghai, China. He was a visiting scholar with the New Jersey Institute of Technology, Newark, NJ, USA, from 2011 to 2012. His research interests include multimedia security, intelligent information processing, cyber information security. Dr. Jiang is an IEEE Senior member.